

Project Report: Employee Retention Analysis

Problem Statement

The objective of this analysis is to understand the factors affecting employee retention and provide insights that can help reduce employee turnover and improve organizational policies.

Methodology

- **Dataset:** Employee records from Kaggle, including satisfaction, evaluation, tenure, department, salary, etc.
 - **Tools & Libraries:** Python, Pandas, NumPy, Seaborn, Matplotlib
 - **Steps:**
 - Data loading and inspection
 - Handling missing values and data types
 - Exploratory Data Analysis (EDA) through statistics and visualizations
 - Correlation and feature analysis
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Key Insights from Analysis

1. Satisfaction Level vs. Retention

Insight: Employees with lower satisfaction scores are more likely to leave. A satisfaction score below ~0.5 significantly increases attrition risk.

2. Average Monthly Hours

Insight: There's a U-shaped relationship. Both overworked and underutilized employees are more prone to leave. Employees working around 200 hours/month tend to stay longer.

3. Salary & Department

Insight: Low salary tiers see higher attrition. Some departments (e.g., sales and technical roles) have higher turnover rates than others like HR or management.

4. Last Evaluation

Insight: Surprisingly, both very low and very high evaluation scores are correlated with employee departure — possibly due to dissatisfaction or burnout.

5. Promotion in Last 5 Years

Insight: Employees who were recently promoted are more likely to stay, indicating the importance of career progression in retention.

Summary

This analysis reveals that **satisfaction level, workload, salary, and promotions** are key factors influencing employee retention. Proactive engagement, work-life balance, recognition, and fair compensation are essential to retaining talent.